

Project Synopsis

1. Title

Working Project Title

AI DSA Tutor – An AI-Powered Data Structures and Algorithms Learning Assistant

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
Deeksha V	GSSSIETW	AIDS	+919380494733	deeksha31005@gmail.com	Lead
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Program Track and Domain

Program Track	Domain
Engg-Spoke	Machine Learning

Guide Details

Project Guide: Gunjan Mishra

Designation: [Subject Matter Expert (AI&ML)]

2. Problem Statement

Data Structures and Algorithms (DSA) are fundamental concepts in computer science and play an important role in programming, technical interviews, and software development. However, students often face difficulties in selecting problems according to their skill level, understanding where they are getting stuck, and knowing how much assistance they need while solving a problem. Traditional DSA practice platforms generally provide problems and solutions but may not provide personalized guidance

based on the learner's difficulty level. To address this gap, **AI DSA Tutor** is proposed as an AI-powered learning platform that provides DSA problems across multiple topics and difficulty levels, offers progressive hints, displays complexity analysis, and predicts the difficulty of problems using a Machine Learning model. The system aims to make DSA practice more structured, interactive, and learner-friendly.

3. Objectives

The main objectives of the **AI DSA Tutor** are:

1. To develop an interactive platform for practicing **Data Structures and Algorithms** problems.
2. To provide DSA problems from different topics such as **Arrays, Binary Trees, Dynamic Programming, Graphs, Hashing, Linked Lists, Recursion**, and other related topics.
3. To classify problems into **Easy, Medium, and Hard** difficulty levels.
4. To implement **Machine Learning-based difficulty prediction** for DSA problems.
5. To provide **progressive hints** that help students understand and solve problems without immediately revealing the complete solution.
6. To display **time and space complexity analysis** for the selected problem.
7. To provide an easy-to-use web interface through which students can select topics and practice problems.
8. To help learners improve their problem-solving skills through guided and structured DSA practice.
9. To demonstrate the practical application of **Machine Learning in an educational domain**.

4. Methodology / Implementation Plan

The proposed system will be developed as an interactive web-based DSA learning platform using Python and Machine Learning techniques.

Overall Workflow

DSA Problem Dataset → Feature Preparation → ML Model Training → Difficulty Prediction → Problem Selection → Progressive Hints → Complexity Analysis → Interactive Learning

1. Dataset Preparation

A dataset containing DSA problems will be prepared with information such as:

Problem statement

DSA topic

Difficulty level

Relevant problem characteristics/features

Hints

Example input and output

Time complexity

Space complexity

The current application contains 30 DSA problems covering 11 different topics, with the problems distributed across Easy, Medium, and Hard categories.

2. Feature Preparation

Relevant characteristics of the DSA problems will be converted into numerical/categorical features that can be provided to the Machine Learning model.

The application currently uses 5 features for the difficulty prediction component.

3. Machine Learning Model

A Decision Tree Classifier is used as the Machine Learning model for predicting the difficulty level of DSA problems.

The model classifies problems into:

Easy

Medium

Hard

The application also displays the predicted difficulty along with an AI confidence score.

4. DSA Problem Practice

Students can select a DSA topic and choose a problem from the available problem set. The platform displays the problem statement along with an example input and expected output.

5. Progressive Hint System

Instead of immediately providing a complete solution, the platform provides progressive hints. Students can reveal hints step-by-step whenever they require assistance.

This encourages students to attempt the problem independently while still receiving guidance when needed.

6. Complexity Analysis

For each problem, the platform displays its expected:

Time Complexity

Space Complexity

This helps students understand not only how to solve a problem but also how efficiently the solution operates.

7. Web Application and Deployment

The application is developed using Streamlit, providing an interactive interface for DSA practice. Python-based libraries are used for data handling and Machine Learning implementation.

Technologies and Tools

Programming Language: Python

Web Framework: Streamlit

Data Processing: Pandas

Machine Learning: Scikit-learn

Machine Learning Algorithm: Decision Tree

Development Environment: VS Code / Jupyter Notebook

Deployment: Streamlit-based web deployment

5. Resources and Timeline

A. Hardware & Software Requirements

- Hardware Requirements
- Laptop/Desktop computer
- Minimum 4 GB RAM
- Minimum 10 GB available storage
- Stable internet connection for deployment and accessing the web application
- Software Requirements
- Operating System: Windows / Linux / macOS
- Python 3.x
- Streamlit

- Pandas
- Scikit-learn
- Jupyter Notebook / VS Code
- Modern web browser such as Google Chrome, Microsoft Edge, or Firefox.

B. Dataset

- A dataset of DSA problems is available for the project.
- The current dataset/application consists of:
 - 30 DSA problems
 - 11 DSA topics
 - 3 difficulty levels: Easy, Medium, and Hard
 - 5 Machine Learning features
 - Problem statements with examples
 - Progressive hints
 - Time and space complexity information
- The dataset will be curated/updated as required to improve the coverage and performance of the application.
- Dataset Source: Project-specific/curated DSA problem dataset.
- Whether dataset is available
- Source link, if available
- If not available, how they will collect or generate it

C. Project Timeline

Phase	Task	Tentative Completion
Day 1	Problem identification, requirement analysis, and project planning	Day 1
Day 2	DSA dataset collection, preparation, and feature selection	Day 2
Day 3	Data preprocessing and Decision Tree model development	Day 3
Day 4	Development of DSA problem practice module and topic selection	Day 4
Day 5	Implementation of progressive hints and complexity analysis	Day 5
Day 6	Streamlit integration, UI development, testing, and debugging	Day 6
Day 7	Final testing, improvements, deployment, and documentation	Day 7

6. Expected Outcomes / Conclusion

The proposed AI DSA Tutor is expected to provide a functional and interactive platform for learning and practicing Data Structures and Algorithms. The system will allow students to select DSA topics, practice problems based on different difficulty levels, receive progressive hints, and understand the time and space complexity of problems. The Machine Learning component using a Decision Tree model will predict the difficulty level of DSA problems and provide an associated confidence score. The final outcome will be a working web-based prototype that demonstrates how Machine Learning can be integrated with educational applications to provide a more structured and personalized learning experience for DSA students.

7. References

1. Python Documentation – Python programming language and libraries.
2. Pandas Documentation – Data manipulation and analysis.
3. Scikit-learn Documentation – Machine Learning algorithms and Decision Tree implementation.
4. Streamlit Documentation – Development and deployment of interactive Python web applications.
5. Standard Data Structures and Algorithms learning resources for problem statements, complexity analysis, and DSA concepts.
6. Relevant Machine Learning and Decision Tree learning resources used during the development of the project.